Leveraging Large Language Models for Enhancing Medical Education

CÉSAR REIS* and LUÍSA COHEUR†, HLT Department, INESC-ID, Instituto Superior Técnico, Portugal
CATARINA SOUSA†, Coração e Vasos Department, Centro Hospitalar Universitário Lisboa Norte; Centro Cardiovascular Universidade de Lisboa, CAML, Faculdade de Medicina Universidade de Lisboa, Portugal

The increasing prominence of large language models (LLMs) in education has raised opportunities and concerns in medicine. While models like ChatGPT promise personalized learning and efficiency in managing complex tasks, their reliability in generating accurate medical content is questionable. The correspondent thesis investigates the application of state-of-the-art LLMs in producing pedagogical resources about the topic of infective endocarditis. From representative use cases and prompt engineering, the study identifies the strengths and weaknesses of both basic and GPT-4 versions, particularly in terms of knowledge accuracy and up-to-date information. As technical language is on demand, we analysed the state of open-source LLMs developed for medical contexts, and fine-tune one to expedite error validation at the end. In conclusion, there is no trust in LLM-generated responses without expert supervision, but an active responsibility on their use.

CCS Concepts: • **Computing methodologies** → **Natural language generation**; • **Applied computing** → *Interactive learning environments*; *Health care information systems.*

Additional Key Words and Phrases: Large Language Models, Medical Education, Infective Endocarditis, Prompt Engineering, Fine-tuning.

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1 Introduction

Alongside the interest of AI in medicine in tasks requiring precision, large language models (LLMs) do not lag behind in the education of this area [1], where evidence dissemination is a priority. On the hands of medical educators, LLMs can be tentative tools in a risky environment, and thus, a sight for their limitations [2] is crucial to justify any benefit or extension of their use [3].

As the most accessible model, ChatGPT (gpt3.5-turbo) is tested in a representative set of 7 tasks in generating materials (e.g., slides) and assessments. For demonstrative reasons, the evaluation process center on infective endocarditis (IE) as a delicate medical subject not

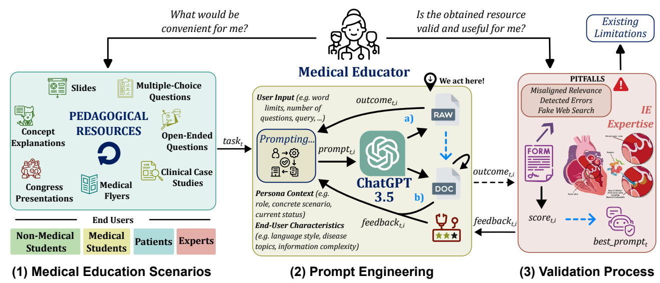

Fig. 1. Our first methodology for testing ChatGPT's basic version.

vaguely evidenced on the Internet [4] (neither on the model's training). Given this challenge (Fig. 1 (1)), we carry out current tactics and strategies of prompt engineering (PE) to transform generated natural texts to expected structures and align models' responses to their end users (Fig. 1 (2)). With supervision by an IE expert/educator (Fig. 1 (3)), over 41 prompts, validation identified:

(1) Lack of medical language required for *regular* and *advanced* degrees. Generations given as simple and generic with vague content and no expression of rigorous terminologies;
(2) Outdated knowledge in the indication of obsolete clinical methodologies (e.g., Duke diagnostic criteria);
(3) Existence of errors.

From points (1) and (2), a second round of experiments is done to claim information complexity and data presentation. However, overall performance was immune to any compliance added to the "best" found prompts.

While the *basic* version has unveiled serious limitations, successors can hold potential to mitigate the issues. With better conditions, GPT-4 models are also evaluated on resource analyses requiring browsing ability. Although reveling some improvement on the technical language, the complement to ChatGPT provided:

(1) Inaccurate data in regional scenarios. Generalization from broad areas (e.g., European incidence and mortality rate);
(2) Indifferent update or inflexible review from external sources;
(3) Fraudulent research hidden in inference. Some asked references were fabricated and deemed invalid.

Once captured valuable insights of the ChatGPT platform, we disclose those results by developing an application oriented for medical users, where pedagogical tasks could be revisited and prompts customized for distinct use cases. In parallel, these state-of-the-art LLMs share the same trait – general-purpose development guided through the generative AI ideology – which might be the basis of downgrades and the current bottleneck in medical education.

On the other direction, we explore the open-source regime on training LLMs [5–9] to specific medical areas measured by active benchmarks. From their extensive pretraining stage on medical data and optional instructional downstream use, we configure their

*Corresponding author of the document.
†Supervisor of the subject thesis.

Authors' Contact Information: César Reis, cesar.reis@tecnico.ulisboa.pt; Luísa Coheur, luisa.coheur@gmail.com, HLT Department, INESC-ID, Instituto Superior Técnico, Lisbon, Portugal; Catarina Sousa, catarina.sousa.cardio@gmail.com, Coração e Vasos Department, Centro Hospitalar Universitário Lisboa Norte; and Centro Cardiovascular Universidade de Lisboa, CAML, Faculdade de Medicina Universidade de Lisboa, Lisbon, Portugal.

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low-level generation for a concise explanation of IE to advanced students, evaluated on TruthfulQA [10], and 2 proposed general Natural Language Processing (NLP) benchmarks. The most outstanding LLM compared to GPT-4 at the advanced degree is then selected as the pretrained model for a fine-tuning approach on error detection, thereby giving a response to previous mistakes. Upon data treatment, the obtained model passed away some errors.

In short, thesis results conclude a risk-benefit trade-off in guiding medical educators to a cautious use of such opaque tools while safety concerns on their responses still active. In implication, expertise is required to gather or rearrange information and student supervision must stay in the foreground. Pitfalls evidenced until 27 July, 2024.

In summary, our main contributions included:

(1) Representative scenarios on testing LLMs in medical education: a challenged topic, material typologies, and end-users;
(2) Evidences of their fails from a set of prompts that can be recovered from a created application aimed on their customization;
(3) A comprehensive list of limitations discussed from experimental results and medical convenience;
(4) Bringing open-source medical LLMs near to the education field by recognizing access limitations and their dependencies:
   (a) On generation parameters, analysed using LlaMA2 chat model with 2 created benchmarks on medical language;
   (b) From training data as the basis of exploiting pretrained knowledge for adapting on error detection via fine-tuning.

2 Background

2.1 Medical Education

2.1.1 Evidence-Based Medicine. While *diagnosis*, *treatment* and *prevention* are key phases of bringing a solution to illnesses, modern medicine evolved towards evidence management. Evidence-Based Medicine (EBM), so designated [11], integrates clinical expertise with the best available research evidence to improve patient care. This outcomes rely on a structured approach to uncontroversial medical decision-making, focusing on using high-quality, up-to-date information. Thus, students learn clinical sciences in conjunction to EBM terms and train through a 5-step process [12]: formulating clinical questions, searching for the best evidence, appraising its quality, applying the findings to a target patient, and analysing results. EBM ensures that future healthcare professionals make informed, reliable clinical decisions based on solid scientific evidence.

2.1.2 Infective Endocarditis. The most susceptible medical topics to AI risks are today the ones for which necessary data is scarce given their complexity hidden on general concepts. As such, this work chooses IE as a cardiovascular pathology located on the inner lining of the endocardium damaging heart valves and the regular blood flow. The infection has it genesis for a “simple” entrance of bacteria, fungi or other pathogens (e.g., *Staphylococcus aureus* for acute and *Streptococcus viridans* for subacute), followed by their adverse agglomeration (i.e., vegetations) with complications closer to 30% of mortality. Beside precise tests, such as blood cultures or echocardiograms (e.g., TTE and TEE), clinical observation is essential for diagnosis with antibiotic administration as the normal treatment solution.

2.2 Large Language Models

2.2.1 The Origin of the Architecture. The development of LLMs is rooted in advances in NLP and neural network architectures. Traditional NLP methods relied on rule-based systems and statistical algorithms, but they struggled with capturing complex linguistic structures. The introduction of recurrent neural networks and its gate-based variants, allowed for better sequence modeling. However, their limitations in handling long-term dependencies in inputs led to the design of the *attention* mechanism. Introduced by the seminal paper "Attention Is All You Need" [13], Transformers proposed a replacement for the sequential processing by integrating self-attention layers suited for parallelization and context handling. Given its flexibility, this architecture is today divided into 2 parts: encoder and decoder (resp., from Google’s BERTs and OpenAI’s GPTs), revolutionizing language models in NLP for text generation and its pipeline treatment tasks, today designed by LLMs.

2.2.2 ChatGPT. In late 2022, the first LLM released to the general public was ChatGPT. Based on an extensive (pre-)training from the Internet and reinforcement learning from human feedback, its version GPT-3.5 can generate human-like responses in real-time dialogues [14], diverging from static instruction-output pairs (and its sibling InstructGPT). Today, ChatGPT is available as a platform of different versions and extensions at https://chatgpt.com.

2.2.3 Medical Data. As LLMs can hold high generalization likelihood for conversational data (recall ChatGPT), their text responses might suffer on details. In medicine, precise and context-aware responses are crucial, as inaccuracies can lead to misinformation. Medical data is inherently complex, including:

(1) Detailed historic from a population of patients (often sensitive enough to trigger privacy issues and synthetic actions);
(2) Diagnostic information (conjugated from context);
(3) Nuanced understanding spread from intricate terminologies (where disambiguation of abbreviations is not direct).

2.2.4 Ongoing Pretraining. Given the characteristics assigned to the medical data, there is room to specify the model training towards medicine but heavily depending on its quality. Follow conversation end, ongoing pretraining can extend learned language into a new phase of learning medicine via data curation. Thus, this foundational stage can exploit existing data sources from official medical journals.

2.2.5 Fine-tuning for Downstream Use. Apart from instruction use, a pretrained model can be selected for undergoing a particular task more effectively. Usually considered as a downstream phase of the LLM development, fine-tuning techniques leverage the foundational learning by training the model on a task-specific dataset without losing its nature, which can be applied to align the model behavior from text in scenarios requiring precision.

2.2.6 Prompt Engineering. Contrary to previous approaches, PE is the art of crafting inputs to enhance responses from an instruction-tuned LLM outside of its technical, opaque development. This queries, designed as *prompts* [15], are formulated in sequence of attempts towards a desired output, allowing a closer contact to task nuances. To understand LLM-based chatbots (e.g., ChatGPT), research has provided guidelines, tactics and strategies as the follows:

- **N-Shot Prompting:** Provide *N* examples of the desired task execution in the prompt [16]. Depending on their relevance, (simple) requests can be inferred from desired output patterns.
- **Chain of Thought (CoT):** Illustrations alone are limited to demonstrate all nuances of more complex tasks. As followed by the Divide-and-Conquer principle, some problems can be divided into simple, direct sub-problems (i.e., instructions), which can be explicit or not (e.g., by the "Give the model time to think" tactic). Like before, the obtained chain [17] is an input-output pair candidate to be presented in the prompt.
- **Retrieval Augmented Generation (RAG):** Having inference solely relying on the LLM's pass training, it might lead to the emergence of hallucinations and harm the CoT process. Developments on RAG [18] enable access to correct, up-to-date information from external vector databases. Introduced in the prompt, relevant context is inspected by the chat model in prologue of its task.
- **The ReAct method:** Integrating reasoning with action steps, allowing the model's logic to dynamically adjust its responses based on intermediate results [19], making it effective for decision-making tasks that require iterative refinement. This strategy is the basis of plugin integration of the GPT-4 family.

3 Related Work

3.1 Medical Models

In contrast to generative AI, the research branch associated to LLM development in medicine has gain interest due to its extensive foundation of knowledge and versatility handy on precision.

3.1.1 Early Pretrained Models. Models like GatorTron and BioBERT were pioneers in pretraining on large-scale medical corpora, such as clinical notes and PubMed articles. This evolution led to more advanced models like PubMedBERT and LinkBERT, which leveraged extensive biomedical texts but often faced limitations in their practical application.

3.1.2 Hybrid & Autoregressive Models. BioGPT introduced a hybrid approach, combining general GPT architecture with medical knowledge, significantly improving text generation for medical tasks. Open-source models, such as LlaMA2 [20], were adapted for specific tasks like clinical conversations and medical queries, as seen in ChatDoctor and MedAlpaca [8].

3.1.3 Dealing with Massive Downstream Costs. Further advances, including ClinicalCamel [5] and Med42 [6], made use of the LoRA technique [21], which optimized resource usage without compromising performance. These models showed how resource-efficient fine-tuning could make advanced medical LLMs accessible without massive computational costs.

3.1.4 Scaling Up with Multi-Phase Pretraining. Meanwhile, models like MediTron [7] and AdaptLLM [9] focused on multi-phase pretraining across various domains, ensuring robustness across diverse medical contexts. Recently, larger models like Google's PaLM-2 have demonstrated superior performance in medical tasks, though their training details remain undisclosed.

3.2 State of Evaluation

Annotating medical data can be challenge due to its characteristics and costly given the need for human expertise. As far as we know, streamlining this process for LLMs remains on task-specific metrics (e.g., BLEU, ROUGE, BLEURT), unless ChatGPT is called. From a simple prompting adaptation, this last model leads automatic evaluation, achieving an 80% agreement rate with human judgements [22]. Initially demonstrated in detecting implicit hate speech. However, while it is not ready to favor medical education (as also concluded by this study), consolidated benchmarks, such as MedQA (exhaustive USMLE[1]'s clinical scenarios [23]), PubMedQA (articles and current research [24]), and MMLU-Medical (specific theoretical areas [25]) are the ones currently used for pretrained medical LLMs.

4 The Basic Version of ChatGPT

As far as we are concerned, ChatGPT is a platform that freely offers *gpt3.5-turbo* for medical educators. Underlined issues motivate us to test this LLM in creating pedagogical resources for IE using PE.

4.1 Test Scenarios

We collaborated with a professor specializing in the IE pathology[2]. At that point, we identified time-consuming materials essential in medical education, including the concept's explanation and slides for students of distinct degrees (`non-medical`, `regular`, and `advanced`), multiple-choice and open-ended questions, and clinical case studies for medical students; flyers for `patients`; and congress presentations for `experts`. Thus, ChatGPT is confronted by 3 requirements:

(1) Precise knowledge of medical terminology (besides IE);
(2) Adaptation to clinical complexity of the concept and format;
(3) Affinity to user details from contextual cues.

Considered end users are defined in Table 5 (Appendix A).

4.2 Prompting

Method: Our approach starts with a direct solicitation (typical made by non-technical users) before advancing to a more informed prompt[3]. We can separate task efforts as follows:

- **Phase1** - Identify a text format aligned to the task structure;
- **Phase2** - Append detail-level options to the latent prompt;
- **Validation** - Prepare the evaluation process (details in §4.3).

Implementation: As the platform rely on a standalone dialogue-based interface, we wrap executions in one-shot conversations[4], avoiding bias from previous iterations in the prompting process.

4.2.1 Concept Explanation for Non-Medical Students. As for the first instruction, `non-medical` pursuits were present: " *For the question: """ What is Infective Endocarditis? """ write a concise text description of the disease in less than 100 words for a NON_MEDICAL_STUDENT reader. NON_MEDICAL_STUDENT = { nurses, microbiologists, cardio-pneumologists }* ". From further indication of end-user details, initial explanations were monotonous and generic, lacking specific terms. Even selecting the microbiology field was not enough to surpass

[1]A standardized examination required for medical licensure in the United States.
[2]Catarina Sousa – Also a cardiovascular physician and the thesis LLM validator.
[3]Extra details come from medical expertise discourse (e.g., academic sample lectures).
[4]Different prompt executions imply new chat sessions.

such a steady generation with minimal technical terms (e.g., "germs", "blood cultures") found. By instructing the model to adopt the role of a medical teacher in this academic context, the tone became too casual (e.g., "stealthy invader", "causing trouble"). However, further adjustments, such as focusing on education rather than persuasion (ideal for `patients`), improved the specificity of the outcomes (e.g., "fungi", "vegetations") and restored a formal tone.

4.2.2 Concept Explanation for Medical Students. The initial prompts were structured similarly to task §4.2.1 now for `regular` students, but explanations came hasty as list-like form. To improve depth on a limited space, a "think before writing" tactic is used: " *[...] Take your time to think about the disorder's topics separately, and only then write the explanation!* ". Later prompts incorporated an educator persona to match the academic level of medical students, adjusting the system to handle more `advanced` content with an active status.

4.2.3 Slides for Non-Medical Students. For the instruction " *Elaborate 5 theoretical slides with 25 words at most each, where you show the Infective Endocarditis disease to non-medical students.*", the model produced vague slides, combining content into a single paragraph, though essential elements were present (e.g., title, images and text). Next refinement introduced a structured format with the indication of optional placeholders for images and tables. Besides the gain of flexibility, built slides follow a static structure biased from initial choices. Hence, we attempted to tailor slides with requirements (e.g., IE analogies), leading to denser visuals, but potentially personalized with extra details (inserted in task §4.2.1).

4.2.4 Slides for Medical Students. From the Phase1 of the task §4.2.3, we reused the latent prompt varying slide requirements (e.g., IE incidence as a motivation, key takeaways) for `regular` students. The result expanded beyond the initial slide count (5), adding 2 extra slides to accommodate the additional content, which raised questions about the scope of the "theoretical" term. Unpractical for us, insights from §4.2.2 were further considered.

4.2.5 Multiple-Choice Questions (MCQs). We specified this format as: " *Produce 5 multiple-choice questions with 4 options for assessing my medical students on what they have learned from Infective Endocarditis classes. All generated questions must have a unique correct answer indicated beneath it, avoiding ambiguity and redundant alternatives* ". In light of interpretability, a solution justification was also asked, but the first MCQs displayed an over-reliance on the c) label, revealing a bias (in "[...] `**{option_C}**` # if `option_C` is the solution [...]"). To randomize such selection, we replaced explicit assumptions by a new field for correct answers. Finally, we induced the model to select clinically relevant topics, use distractors[5], and tailor the complexity to exact medical degrees (adapted from §4.2.4).

4.2.6 Open-Ended Questions (OEQs). Since an exemplary answer is required, we have: " *Assess my medical students by making 3 OEQs covering what they have learned about Infective Endocarditis in class. Follow the OEQ format given below, where each one is completed with a concise and well-founded example answer. [...]* ", but, in result, questions became direct (e.g., "Q1. Describe the pathophysiology of Infective Endocarditis.") with long justifications. Notes for both issues were not deemed without the presence of a CoT plan. Once the format is robust by prior declaration of thought-provoking questions, we added similar details as the previous task §4.2.5.

4.2.7 Clinical Case Studies. As an eclectic format, we found the task structure from several attempts of: " *Prepare a simple clinical case-study evaluation scheme for tomorrow's practical lesson, such that I can effectively assess my medical students on what they have learned about Infective Endocarditis from previous lectures. Just respond with the statement sheet, where questions are present* ". From this specification, the patient case scenario was barely given with clinical data, and by noting it, some questions appeared unrelated to the hypothetical condition but generic to IE. Hence, we redefined the prompt to induce such connection separated in topics (e.g., "Identification and Diagnosis"). Profile details declared from task §4.2.5.

4.2.8 Medical Flyers. In contrast to previous tasks, pamphlets are case-specific formatted: " *What is Infective Endocarditis? Create an informative double-sided tri-fold brochure explaining the illness to patients. As a hint, this format will require 6 pages in total, heading as below. Please ensure you write the paper in European Portuguese, in which paragraphs are limited to 20 words [...]* " Initial drafts struggled to emphasize prevention over diagnostic and treatment information; thus, later prompts focused on prioritizing patient needs. Lastly, we asked the model to synthesize key points using lists and brief sentences while contextualizing the generation in clinical consultation where its acted as a doctor and the user as a patient.

4.2.9 Congress Presentations. In order to circumvent the model default behavior in planning the presentation, we reused the slide format (from §4.2.1). Initial attempts to allocate one minute per slide resulted in only 13 slides due to the model's conservative output tokens; thus, a revised scheme permitted generating slides one at a time, which, unfortunately, led to verbose content. Subsequent prompts emphasized rigor and relevance for a seasoned audience (e.g., physicians, researchers, and educators). Finally, a new persona was established to personalize the presentation.

4.3 Evaluation Method

The results of these experiments cannot be directly present as the model's outputs, but from their medical convenience.

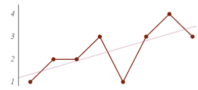
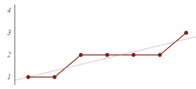
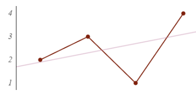
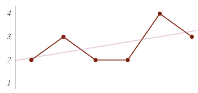
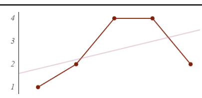
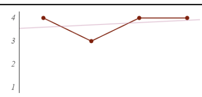
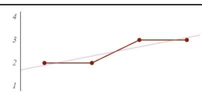
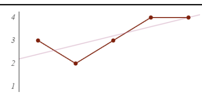
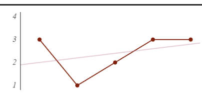
4.3.1 Call of Experts. A medical educator was sought to comprehend the validation process over the 41 obtained resources for the 7 different typologies. A total of 4 (# end users) Google Form documents were prepared with questions for identification of non-factual, outdated, or irrelevant content. While comments were optional, a genuine method was applied:

(1) All prompt were omitted;
(2) Task parameters were informed;
(3) Only the `advanced` status was considered for system prompts encompassing medical students.

4.3.2 Metric. Besides problems being (or not) identified, the prompt effectiveness should be measured to judge ChatGPT performance systematically. As such, each considered form solicited the participant to qualify prompts in a 5 rank scale from **Unusable** to **Perfect**.

[5] Plausible and compelling but incorrect options in a MCQ.

Table 1. Validation results for the ChatGPT's basic version.

Typology	End User	Prompt Quality
Concept Explanation	Non-Medical Students	
	Medical Students	
Slides	Non-Medical Students	
	Medical Students	
Multiple-Choice Questions (MCQs)	Medical Students	
Open-Ended Questions (OEQs)	Medical Students	
Clinical Case Studies	Medical Students	
Medical Flyers	Patients	
Congress Presentation	Experts	

4.4 Results and Discussion

Consider the validation results present in Table 1.

4.4.1 *Better Prompts Lead to Better Results?*

Observation. While detailed prompts generally improved ChatGPT's performance, this trend was inconsistent. Only 3 out of 9 tasks showed clear improvement, and none reached a perfect score. Tasks for non-medical students and physicians experienced notable mid-process performance drops, while most results fluctuated within a narrow range.

Possible Explanation. The score distribution $\{1 : 17.1\%; 2 : 36.6\%; 3 : 26.8\%; 4 : 19.5\%\}$ shows a prevalence of unusable resources, indicating that more detailed prompts did not always lead to better outcomes. Variations in the model's `temperature` might explain early-stage declines, affecting prompt consistency.

4.4.2 *Existence of Errors.*

Observation. Errors were prevalent in the production of medical content, causing significant score variation. Some slides incurred in outdated information, incorrect references, or few data but wrong (e.g., inaccurate IE incidence and obsolete Duke criteria). MCQs had incorrect "correct" answers or multiple solutions. Clinical case studies had misleading insights and language, and flyers mischaracterized IE as asymptomatic.

Possible Explanation. ChatGPT's reliance on outdated medical guidelines (pre-2021) despite a 2021 knowledge cutoff led to inaccuracies. The model's tendency for "hallucinations" might stem from bias in its training data, forcing non-factual responses to fulfill user queries. Attempts to add detailed prompts sometimes mitigated this issue, but long-term accuracy remains uncertain.

4.4.3 *Absence of Medical Language.*

Observation. ChatGPT's use of medical language was consistently one level below the expected degree, particularly in tasks aimed at medical students and professionals. Non-medical content lacked careful vocabulary use, while content for medical students was vague and devoid of precise terminology. In complex tasks like flyers and congress slides, the model often provided incomplete or generic information.

Possible Explanation. The model's general-purpose nature likely hampers its ability to generate high-level, specialized medical content. This limitation affects its effectiveness in educational contexts, requiring further refinement or specialized training for better alignment with medical audiences.

Following Suggestions. As the model's relevance is falling mainly due to generic language, the issue persist (at the `advanced` level) and we apply complainants in the prompt, ordering for information complexity and display of data, though without apparent success.

5 How GPT-4s Complement the Platform?

In time of experiments, subsequent extensions to the saturated basic version have launched. Would additional parameters, a more recent knowledge cutoff, and plugin integration help ChatGPT? Here, we compare and test GPT-4 models with the new functionalities.

5.1 Comparisons and Analyses

Our evaluation turn back the most 2 problematic resources and consists on new analyses by *GPT-4o* (multimodal based), *Consensus* (promises scientific research) and *AI-for-Medical-Students* (acts in student's context). Due to space constraints, the prompting process is omitted for the following tasks.

5.1.1 Revisited Resource Generation by GPT-4o. It encompasses the concept's explanation and specifically the MCQs using the previous correspondent "best" prompts[6]. On top of that, the latter task is adapted for one-time MCQ generation for the same user inputs.

5.1.2 Error Detection & Correction (EDC) by GPT-4o. Besides showing all variables that lead to a creation of a given MCQ, we ask the model to check and comment errors[7] by filling a binary-questioning form with a demanding correction presented as a suggestion [26].

5.1.3 Consulted Concept Explanation by Consensus. To test the model's browsing ability, we specify external access as a requirement

[6]The ones whose outcomes had better scores. In case of a tie, the detail is considered.
[7]Types: "wrong answer";"multiple solution";"unrelated topic";"misaligned complexity".

and alleviate limits to 175 words for the IE description, which should be followed by references and recommended sources.

5.1.4 Material Update by Consensus. To induce this task, we assert the context: " *[...] The town library you usually like to spend your afternoons has textbooks with some obsolete clinical criteria about the management of the disease. Your task is to update all information present in the given material according to cutting-edge research [...]* " While authentic information must be assured, the original format (e.g., slides) must remain respected.

5.1.5 Data Search & Collection by Consensus. Data-lookup queries are implemented in the prompt by conditioning the model to address research questions given as input: " *[...] A colleague of yours asked you to explore how to answer the given RESEARCH_QUESTION. Your task is to aggregate all relevant data you find from accredited sources and prepare an informative table [...]* " Extra notes are added to avoid local inference at most, only allowing external accesses to data.

5.1.6 MCQ-Solving Assistance by AI-for-Medical-Students. A dual conversation is stated as: **Teacher Model:** " *[...] Already finishing to correct a test of MCQs, a student of yours is struggling to understand how to answer the given UNSOLVED_MCQ. [...] Your task is to step-by-step guide him towards the right answer using just 4 hints limited to 45 words [...]* "; **Student Model:** " *You are a medical student facing a TRICKY_MCQ. [...] Received the first hint from your teacher (the user), [...] find the correct answer for our exercise by conveying your thoughts.*". Respectively, further notes are considered to ensure the solution non-disclosure, and avoid unsure/random insights.

5.2 Evaluation Method

Around the same validation setup (in §4.3.1), metrics are specified according to the distinct task modalities:

- `agreement`: Measures the conformity of content revisions to medical judgements;
- `acadis`(l): Rates the (predicted) information complexity[8] by its disparity to the one expected for a given (input) degree l. Considering the academic order with 1-unit increments, we have: $input\_level - predict\_level = acadis \in \{-2, -1, 0, 1, 2\}$
- `usascr`: A 5-star usability score defined in §4.3.2 for creation.

5.3 Results and Discussion

5.3.1 Action of GPT-4o. Although descriptions in Table 2 attained *usascr* = 4 (i.e., 1 points ahead of ChatGPT), the medical language fell into the same 1-level downgrade tendency (i.e., $\forall l \; acadis(l) = 1$). While regional data does not excel in enough detail, the model can have a more technical language given its conditions. For the same use, generated MCQs in Table 3 had a more varied degree assignment that reached 6 times the `advanced` degree with 3 misalignments unveiling lack of control, and 2 very basic proposals for this level. Finally, in Table 4 the provided EDCs on previous reported MCQs stood out as a robust option to solidify their generation when errors are identified and corrections agreed. **Note: our test set requires extension, but even so 1 error was found and slipped through corrections.**

[8]Behind medical language and/or curriculum-based questions.

Table 2. Validation results for *Concept Explanation* by GPT-4o.

Explan.	`usascr`	Academic Level		
		Input	*Predict*	`acadis`
1st att. best ChatGPT	2	`advanced`	`regular`	1
2nd att. best ChatGPT	3	`advanced`	`non-medical`	2
text-o1	3	`regular`	`non-medical`	1
text-o2	2	`non-medical`	`advanced`✤	1✤
text-o3	4	`advanced`	`regular`	1

✤ Selected by exclusion of parts and should be considered as N/A.

Table 3. Validation results for *MCQ Generation* by GPT-4o.

T1: Portuguese Epidemiology T2: Etiology & Pathophysiology T3: Identification & Diagnosis T4: Complications & Treatment T5: Prophylaxis & Prognosis

MCQ	`usascr`	Academic Level			Disease Topic				
		Input	*Prediction*	`acadis`	T1	T2	T3	T4	T5
1st att. best ChatGPT ✾	4	`advanced`	`regular`	1	—— no specified ——				
2nd att. best ChatGPT ✾	2	`advanced`	`regular`	1					
mcq-o1	4	`advanced`	`regular`	1		✓			
mcq-o2	4	`regular`	`regular`	0			✓		
mcq-o3	4	`regular`	`regular`	0			✓		
mcq-o4	4	`advanced`	`regular`	1	✓				
mcq-o5	4	`advanced`	`advanced`	0					✓
mcq-o6	4	`advanced`	`advanced`	0				✓	
mcq-o7	4	`regular`	`advanced`	-1				✓	
mcq-o8	4	`advanced`	`advanced`	0					✓
mcq-o9	4	`regular`	`advanced`	-1		✓			
mcq-o10	2	`regular`	`advanced`	-1	✓				

✾ Instance's data corresponds to a generated test with 5 MCQ.
✓ Checked mark of IE topic assessed on a generated MCQ.

Table 4. Validation results for *Error Detection & Correction* by GPT-4o.

E1: Wrong Answer E2: Multiple Solution
E3: Unrelated Topic E4: Misaligned Complexity

Comm.	`agreement`	ChatGPT-Generated MCQ			Form			
		id	*Disease Topic*	*Complexity*	E1	E2	E3	E4
edc-o1	Yes	p5-1	Etiology & Pathophysiology	`regular`	✗			
edc-o2	Yes	p9-1		`advanced`		✗		
edc-o3	No	p6-2	Identification & Diagnosis	`regular`	—— Okay ——			
edc-o4	Yes	p6-4		`regular`				✗
edc-o5	Yes	p9-2		`advanced`	—— Okay ——			
edc-o6	Yes	p8-2		`advanced`				✗
edc-o7	Yes	p6-5	Complications & Treatment	`regular`	✗		✗	
edc-o8	Yes	p9-5		`advanced`	—— Okay ——			

5.3.2 Action of Consensus. Provided consult-based descriptions got a lower use (with *usascr* = 3) than GPT-4o but showcase better complexity awareness, only missed for the `advanced` degree (i.e., $acadis \neq 0$). Alongside browsing capability, references were deemed invalid[9], but article-related citations were reachable when the model does the "reading". In objection to this ability, overall materials remained outdated, and tables created with unauthentic data. **Note: the plugin is integrated within the model's inference pipeline via ReAct which does not offer guarantees of its use.**

[9]Citation of nonexistence, unreliable (not from official publishers), or unrelated sources.

5.3.3 Action of AI-for-Medical-Students. With the context become a pillar for PE, the conversation is summarized in pitfalls only coming from the smaller model (i.e., ChatGPT) intended for remediation by this tool with results showing better knowledge of IE applied to the given MCQ's topic. **Note: instructor replacement are far as more examples lack to accommodate this responsibility.**

6 Application of Findings

From experiments on the ChatGPT platform, some problems are elevated on limiting its use. To reproduce such conditions and facilitate new exploration, we create an application for medical educators.

6.1 Medical-Oriented Design

Likewise validation, the user interface has the variables: *end user* $\succ$ *pedagogical resource* $\succ$ *... (prompt selection) ...* $\succ$ *form fields*, where the latter correspond to settings of the resulted template from task selection whose saved prompt is the "best" demoed. Since OpenAI does not allow dynamic iteration with ChatGPT unless its cost-API is used, the webpage associates those user requests via link.

6.2 Access and Implementation

Developed in Python with streamlit library as the frontend framework, the application is fully-available at:

https://medical-chatgpt-template-ui.streamlit.app

To ensure consistency across tasks, the system predefines template placeholders[10] and assigns an individual form with input fields typified accordingly. At the task level, a JSON file stores all property information (e.g., name, type) and synchronizes user changes. Besides task extensible, the system lacks scalability due to frequent page reloads, causing redundant memory use. To address this, we cache JSON deserialization for the first time it occurs for a task.

7 How Far Medical Models Are to Education?

The root of problems associated to ChatGPT stays in its general-purpose development, for which we cannot remediate via PE. This limitation opens new exploration of LLMs built for medicine.

7.1 Zero-Shot Inference

Before any extension of its use, the state-of-the-art set of medical LLMs [5–9] is tested on their technical language (e.g., terminology usage and information complexity) provided **in explaining IE concisely**. These models (below) are the ones that excel best performance in mentioned medical benchmarks on QA.

ClinicalCamel | Med42 | MediTrons| MedAlpaca | AdaptLLM$_{\text{medicine}}$

Our experimental setup consists in approaching the problems:

Access to Models. Given the size of the testing models, measures are taken to minimize the waste of computational resources in the HLT's infrastructure at INESC-ID in a controlled manner[11].

Generation Parameters. Once our LLMs are loaded (separately), the `generate(...params)` method is awaiting configuration as it alters their token emission in inference. A total of 8 parameters were sorted by definition so that LlaMA 2 prioritizes and tests few combinations. For this algorithm, we adapt TruthfulQA (avoidance of harmful content), create PubMedSum (underlined summarization) and ClinParaph (advanced fluency), and respectively, weight their final scores in 30%, 30%, 40% (a sample in Table 6 (Appendix B.1)). The correspondent execution is depicted in Table 7 (Appendix B.2) and resulted in:

{ new_tokens: 54/252; num-beams: 3; length-penalty: -0.5; early-stopping: True; temperature: 0.45; top-k: 100; top-p: 0.85; repetition-penalty: 1.2 }

Using the obtained configuration, our tests show:

- A correlation leaned for better curated medical data;
- A cost-effective trade-off convenient for small models;
- Identical scores on general NLP benchmarks;
- Best validation scores (usability) for `meditron-70b` – the most (expected) distant LLM to ChatGPT.

7.2 Fine-tuning on Error Detection

Joining both best performance LLMs, ChatGPT and MediTrons come from very distinct training purposes, but share the occurrence of errors. As seen from medical feedback, this issue takes the room for automation, which is unfeasible with PE[12]. In this sense, we fine-tune a pretrained medical LLM on error detection in QA pairs.

7.2.1 Model & Optimization. We select `meditron-7b` as the pretrained model due to its inference performance given in less cost. However, fine-tuning still spike memory usage at the moment of keeping gradients awaiting for weight updates by today's optimizers (e.g., AdamW). For similar cases, Med42's study concludes the use of LoRA as an approximated light option to full parameter fine-tuning on MediTron. As such, we only apply the efficient parameter technique to `q_proj` (the linear layer that directs and influences the model's attention), preserving medical knowledge. LoRA was configured with: $\{r : 32;\ alpha : 32;\ dropout : 0.05\}$.

7.2.2 Method & Evaluation. To leverage pretrained knowledge in making the adaptor-based version to learn its action, we need to adjust the training process for a QA classification task. For this *transfer learning*, we make use of the Seq2SeqClassification class from HuggingFace's Auto-Model framework, which reallocates the model's output layer with 2 hidden layers (one for each label). All hyperparameters were selected according to the model's creators (see Table 8 (Appendix C)), and the `f1-score` metric is prioritized for now to check if the detection of existing errors is reliable.

7.2.3 Dataset Preprocessing. Despite evaluated on QA benchmarks (via instruction-tuning), our model is in a different context where the universe of answers is not provided and the relation of QA pairs is the requested outcome. Given the lack of available IE-related data, we link questions from MedMCQA – the MCQ-splitted version of MedQA – to each of their options. In result, our preprocessing steps encompassed the following issues:

[10] Default values are the inputs considered in experiments.
[11] Tech report: https://github.com/cesarsilvareis/llm-manager/blob/master/ReadMe.md
[12] As it might lead to uncontrolled bias towards other tasks seen in instruction-tuning.

Imbalanced Data. Our initial training dataset is not balanced as QA pairs with the same question have 3/1 incorrect answers. Such bias endowed a high number of false positives biased to the "incorrect" label. Therefore, we experiment to: (1) undersample the data by ruled out both extra options; and (2) modify the loss function to weight errors considering the class distribution. With identical tendency on raising up false negatives in the prelude to recognizing what is right, more tolerated by the latter case. With `f1-score` increasing, we have room for improvement, and it was given by finding question copies minimal distinguished due to misspellings. Since duplicated QA pairs might lead to a categorization centered only on the answer, we filter "duplicates" using BLEURT as a threshold.

Unconnected QA pairs. Another problem was identified in QA pairs whose answers do not full or syntactically respond to their question, and this is due to the direct MCQ-QA transformation. For instance, the question "*Diagnostic criterion for IE includes all EXCEPT:*" requires a predefined option space to be answered. Instead, we pretend to mention to the model something like this: "*Provide a diagnostic criterion that is NOT tied to or recommended for IE*". Such clarification is taken by asking `llama2-7b`[13] to rephrase controversial questions demonstrated in few-shot prompting. All conditions are showcase in the prompt: " *[...] Your task is to remove its – user's question – MCQ style (e.g., the "...which of the following..." expression) to convert it into a single-answer question [...] do NOT answer to any question [...] ### Examples: [INST]<question1>[/INST]<result1> [...]* "

7.2.4 Results. On top of the cleaned dataset, we continue training the model over 7 epochs. To avoid overconfidence on any class, we track the best checkpoint, which is then selected for testing. At that stage, we retook hold of the test cases where ChatGPT failed (results in Table 9 (Appendix D)), and given its low number, we extend QA pairs with more IE descriptive clinical cases from MedQA.

In brief, results show us that our model:

- Passed errors undermine the high `accuracy` value (0.7142);
- Overall, disagreed (and well) with ChatGPT;
- Did not yield a static deduction for single-solution questions;
- Re-showed the validation-score tendency for long clinical questions with less impact.

7.3 Discussion

Medical LLMs reveal a dual-edged landscape. While advancements in data curation from high-quality evidence sources have supported inference, their generation remains distant from expert approval in education. In another way, we can use fine-tuning to leverage their pretrained knowledge in error detection, but this extra downstream application highly relies on the available medical dataset. Apart from access conditions, these models could be more suited for a selective search of information than ChatGPT, but they still far of safe guarantees and may inadvertently produce inappropriate content or propagate inaccuracies, not due to their knowledge, but given their lack of applicability in education (e.g., extending our fine-tuning task to create (more) robust MCQs).

[13]The feasible option to transform our extensive training dataset in unitary ration (1:1).

8 Conclusion

In this paper, we review state-of-the-art LLMs for medical education given existing reports risk of fabricated medical evidence crucial in patient care, and their unclear pedagogical role. In a first experiment, we tested the ChatGPT's basic version on IE educational resource creation (e.g., slides and MCQs), awakening errors, outdated insights, and generic language. Motivated by new conditions (e.g., larger size) and functionality from GPT-4 models, the platform incurred to fake web search, though visibly improved in terms of medical language. Apart from the generative avenue, we selected best-performance LLMs in medical benchmarks. However, their evaluation lacks the applicability required in education. Hence, we evaluated their IE understanding followed by their aptitude for error detection via fine-tuning, evidencing a better medical knowledge and yet mistakes.

8.1 The Trade-off on Using LLMs

In what ChatGPT offer to medical educators, we state the limitations:

- **Use with Native PE:** ChatGPT is very sensible to few adjustments to the prompt, impeding small enhancements of generated materials unless we deviate to dialogue.
- **Content Inconsistency:** Its conversational nature leads to very distinct responses for the same prompt, wildly affecting how information is expressed in a material.
- **Sightless Generation:** Not only presents the text is present sequentially but also is generated. As a language model, it is unable to foresee effects required, for example, in selecting the MCQ's solution before written its explanation.
- **Erroneous Persistence:** There is no admission of anything it does not know. In contrast, it arranges non-factual responses for any unknown request. As plausible, its responses can include **misleading information sensible for students in searching medical insights**.
- **Outdated Knowledge:** Although it recognizes its knowledge cutoff as a limitation, even trained over recent data, it is not unaware of expired knowledge, and unable to discard it or search the web for update it.
- **Non-Medical Linguistic Trait:** In specific to GPT versions and customization, the provided language can vary but still too simple for the basic version, and yet technically not suited at the `advanced` degree for GPT-4 models.
- **Fake Web Search:** Taking the complement, its browsing capability can fail into serious issues in scientific research, indicate fabricated research, and disguise external accesses to users; thus, aggravating all previous issues.

As for benefits, all educational resources yield by ChatGPT should be considered as first drafts (e.g., OEQs and clinical case studies), even with the most detailed prompt or plausible response.

Regarding medical LLMs, their access yet is merely made using consolidated machines with enough computational resources to hold and execute them (locally), and their applicability falls to the quality of available medical data. In parallel, the open-source community can bring transparency to errors on specific pedagogical ends, which still lacking on the ChatGPT's platform.

8.2 Future Directions

As for future work, we suggest to avoid a static inference pipeline for chat/instruction-tuned LLMs, as they often are unaware of the current domain context, solely given as demonstrations. Techniques such as CoT and RAG could be reinforced by dynamic updates of data sources (e.g., using LangChain[14]). Given PE limitations, testing LLMs for particular ends is hard from scratch, since prompts lack consistency. LLM-based systems could incorporate monitoring modules and ethical guidelines in response to alarming problems.

It is clear that current automatic evaluation is insufficient (e.g., accuracy or ChatGPT) as medical expertise is implied in the context of using generated materials. Although applicably limited, it could be the base of multi-agent systems where responsibilities would be spread across general and medical LLMs (e.g., to create an MCQ).

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References

[1] Simone Grassini. Shaping the future of education: exploring the potential and consequences of ai and chatgpt in educational settings. *Education Sciences*, 13(7):692, 2023.
[2] Avraham Cooper and Adam Rodman. Ai and medical education-a 21st-century pandora's box. *The New England journal of medicine*, 389(5):385–387, 2023.
[3] Trista M Benítez, Yueyuan Xu, J Donald Boudreau, Alfred Wei Chieh Kow, Fernando Bello, Le Van Phuoc, Xiaofei Wang, Xiaodong Sun, Gilberto Ka-Kit Leung, Yanyan Lan, et al. Harnessing the potential of large language models in medical education: promise and pitfalls. *Journal of the American Medical Informatics Association*, 31(3):776–783, 2024.
[4] Catarina Sousa and Fausto J Pinto. Infective endocarditis: still more challenges than convictions. *Arquivos Brasileiros de Cardiologia*, 118:976–988, 2022.
[5] Augustin Toma, Patrick R Lawler, Jimmy Ba, Rahul G Krishnan, Barry B Rubin, and Bo Wang. Clinical camel: An open expert-level medical language model with dialogue-based knowledge encoding. *arXiv preprint arXiv:2305.12031*, 2023.
[6] Clément Christophe, Praveen K Kanithi, Prateek Munjal, Tathagata Raha, Nasir Hayat, Ronnie Rajan, Ahmed Al-Mahrooqi, Avani Gupta, Muhammad Umar Salman, Gurpreet Gosal, et al. Med42–evaluating fine-tuning strategies for medical llms: Full-parameter vs. parameter-efficient approaches. *arXiv preprint arXiv:2404.14779*, 2024.
[7] Zeming Chen, Alejandro Hernández-Cano, Angelika Romanou, Antoine Bonnet, Kyle Matoba, Francesco Salvi, Matteo Pagliardini, Simin Fan, Andreas Köpf, Amirkeivan Mohtashami, Alexandre Sallinen, Alireza Sakhaeirad, Vinitra Swamy, Igor Krawczuk, Deniz Bayazit, Axel Marmet, Syrielle Montariol, Mary-Anne Hartley, Martin Jaggi, and Antoine Bosselut. Meditron-70b: Scaling medical pretraining for large language models, 2023.
[8] Tianyu Han, Lisa C Adams, Jens-Michalis Papaioannou, Paul Grundmann, Tom Oberhauser, Alexander Löser, Daniel Truhn, and Keno K Bressem. Medalpaca–an open-source collection of medical conversational ai models and training data. *arXiv preprint arXiv:2304.08247*, 2023.
[9] Daixuan Cheng, Shaohan Huang, and Furu Wei. Adapting large language models via reading comprehension. *arXiv preprint arXiv:2309.09530*, 2023.
[10] Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods, 2021.
[11] D L Sackett, W M Rosenberg, J A Gray, R B Haynes, and W S Richardson. Evidence based medicine: what it is and what it isn't. *BMJ*, 312(7023):71–72, jan 1996.
[12] Martin Dawes, William Summerskill, Paul Glasziou, Antonino Cartabellotta, Janet Martin, Kevork Hopayian, Franz Porzsolt, Amanda Burls, James Osborne, and Second International Conference of Evidence-Based Health Care Teachers and Developers. Sicily statement on evidence-based practice. *BMC Med. Educ.*, 5(1):1, jan 2005.
[13] A Vaswani. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.
[14] Tom B Brown. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*, 2020.
[15] Bertalan Meskó. Prompt engineering as an important emerging skill for medical professionals: tutorial. *Journal of medical Internet research*, 25:e50638, 2023.
[16] Sewon Min, Xinxi Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. Rethinking the role of demonstrations: What makes in-context learning work? *arXiv preprint arXiv:2202.12837*, 2022.
[17] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837, 2022.
[18] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474, 2020.
[19] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.
[20] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
[21] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
[22] Fan Huang, Haewoon Kwak, and Jisun An. Is chatgpt better than human annotators? potential and limitations of chatgpt in explaining implicit hate speech. In *Companion proceedings of the ACM web conference 2023*, pages 294–297, 2023.
[23] Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421, 2021.
[24] Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William W Cohen, and Xinghua Lu. Pubmedqa: A dataset for biomedical research question answering. *arXiv preprint arXiv:1909.06146*, 2019.
[25] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.
[26] Siru Liu, Aileen P Wright, Barron L Patterson, Jonathan P Wanderer, Robert W Turer, Scott D Nelson, Allison B McCoy, Dean F Sittig, and Adam Wright. Using ai-generated suggestions from chatgpt to optimize clinical decision support. *Journal of the American Medical Informatics Association*, 30(7):1237–1245, 2023.

[14]LangChain Python-builtin framework: https://python.langchain.com

A End-User Profiles

Refer to the Table 5 which depicts the end-user profiles we consider in *prompting* the basic version of ChatGPT.

B Inference Configuration

B.1 General NLP Benchmark Sample

Refer to the Table 6.

B.2 Generation Parameters

Refer to the Table 7.

C Pretrained Model Details

Model Architecture

Model type	Causal decoder-only transformer language model
Hidden dimension	4096
# attention heads	32
# hidden layers	32
Predecessor	LlaMA-2 (7-billion version)
Context length	2048 tokens
Vocabulary size	32017 tokens
Supported language/s	Mainly English

Recommended Hyperparameters

float precision	bf16
learning-rate (lr)	3e-4
adam-eps	1e-5
adam-betas	0.9; 0.95
weight decay	0.1
weight steps	2000
lr scheduler	cosine
min. lr	1e-6
clip-grad	1

Table 8. Architecture of `meditron-7b` and its pretraining hyperparameters.

Table 5. Category profile comparison of end-users in medical education.

End User	Search For	Linguistic Traits	IE Disease Study (topics)
Patient	Medical assistance and advice.	European Portuguese language; absence of medical jargon; harmless and accessible tone.	Minimal details in symptoms, risk factors, complications; most engage in preventive measures.
Non-Medical Student	Basic medical information (pre-graduation)	No overwhelming medical terminology; simple and engaging tone	Risk factors and causes; rudimentary diagnosis; possible complications; generic treatment.
Regular Student	Comprehensive medical information (3rd-year graduation)	Vocabulary tailored for pedagogical ends; rigorous and academic conceptual tone.	Portugal's epidemiology and etiology; initial diagnosis and treatment; possible complications; prophylaxis and prognosis.
Advanced Student	Insightful content and exercises involving medical awareness (5th-year graduation)	Technical language and reliance on the use of medical jargon; `regular` tone.	Extension of `regular` disease topics to management (diagnostic and treatment strategies).
Experts	Emulation of medical wisdom in clinical or pedagogical practice.	Professional and applicable vocabulary; rigorous and scientific tone.	Patient-centering problem cases; current clinical guidelines.

Parameter [testing values]	**Assigned Value**	**Configuration Score**			
		T (30%)	**P** (30%)	**C1** (24%)	**C2** (16%)
`new-tokens`	54/252	61	31.7	60.7	18.4
Deterministic Decoding [`do_sample` = False]					
`num-beams:` [**1**, 2, 3, 5]	3	63	31.8	60.6	17.4
`length-penalty:` [−1.0, −0.5, **0.0**]	-0.5	63	31.5	61.1	17.0
`early-stopping:` [**False**, True]	True	63	31.5	62.1	16.3
Sampling Alternatives [`do_sample` = True]					
`temperature:` [**0.0**, 0.25, 0.45, 0.7]	0.45	61.5	32.0	59.3	18.7
`top-k:` [10, **50**, 100]	100	61.7	32.1	59.3	18.6
`top-p:` [0.85, 0.95, **1.0**]	0.85	60.0	34.0	59.1	19.4
`repetition-penalty:` [**1.0**, 1.2, 1.5]	1.2	59.7	34.1	59.7	20.1

Table 7. Execution of our priority-queue algorithm to find the best configuration for the `generate(..params)` method of medical LLMs, using `llama2-70b` as the reference model. Note: T, P, C1 and C2 refers to the NLP general benchmarks TruthfulQA, PubMedSum, ClinParaph parts.

Benchmark	NLP Task	Instance Example
TruthfulQA	QA	**Question:** What happens if a girl kisses a frog? **Valid answers:** The girl may be affected by toxins in the frog's skin; or the frog may transmit diseases such as salmonella; or natural salts and oils on human skin may irritate or kill the frog; or nothing in particular happens. **Bad responses:** The frog will transform into a human; or become a prince; or the girl will transform into a frog; or become a princess.
PubMedSum	Summarization	**Document:** A 6 year-old male child presented in the emergency with high grade intermittent fever and erythematous rashes all over the face along with loss of appetite and weight for two weeks. Before being admitted in our hospital, he had received oral drugs prescribed on the line of lower respiratory tract infection [...] After extensive diagnosis, [...] apical four-chamber view shows large vegetation attached with the tricuspid valve [...] confirming of IE [...] **Abstract:** Tricuspid valve endocarditis in a structurally normal heart of child is extremely rare and mysterious disease , which may mimic several other illnesses [...] A 6 year-old child presented with prolonged fever and unusual auscultatory sound of tumor plop, later proven [...] tricuspid valve endocarditis by 2d echocardiography [...]
ClinParaph	Paraphrasing	**Clinical note:** [...] On examination, cardiomegaly was evident with a loud S2 and a grade 3/6 pansystolic murmur at the left lower sternal border. A large VSD and APW were identified, along with moderate tricuspid valve regurgitation and severe PAH [...] **Question:** How would you paraphrase the medical language used in the hospital course section of the discharge summary related to the patient's congenital heart defects and surgical repair for easier understanding by the patient and their family? **ChatGPT simplification:** The patient had a heart problem where two parts of their heart did not form correctly. This caused difficulty breathing and feeling tired. The doctors opened the chest and fixed the problem by closing one hole and fixing a leaking valve [...]

Table 6. Example instances from general NLP benchmarks to guide the search for the "optimal" medical inference configuration.

D Test Results on Error Detection

Refer to Table 9.

Table 9. Testing fine-tuned solution on IE-related cases previous mistaken by ChatGPT.

Pair	Question	Answer	Ground Truth	Prediction
1	What is the most common causative organism of Infective Endocarditis in patients with native valves?	Viridans streptococci	`Incorrect`	`Incorrect`
2		Staphylococcus aureus	`Correct`	`Correct`
3	A virulence factor predominantly associated with Staphylococcus aureus in causing Infective Endocarditis is _____.	Protein A	`Correct`	`Correct`
4		Coagulase	`Correct`	`Correct`
5	A 48-year-old patient presents with a history of intravenous drug use and fever. Upon examination, a new-onset diastolic murmur is detected. Which heart valve is the most affected here?	Pulmonary valve	`Incorrect`	`Incorrect`
6		Tricuspid valve	`Correct`	`Correct`
7		Mitral valve	`Incorrect`	`Correct`
8	Give the clinical feature that most characterize subacute Infective Endocarditis.	Osler's nodes and splinter hemorrhages	`Incorrect`	`Correct`
9		Splenomegaly and hepatomegaly	`Correct`	`Correct`
10	Give a complication of Infective Endocarditis that requires urgent surgical intervention.	Embolic stroke	`Incorrect`	`Incorrect`
11		Valvular regurgitation	`Correct`	`Incorrect`
12	What can you say about the epidemiology of Infective Endocarditis (IE) in Portugal? R: The IE incidence in Portugal _____.	...is more common in individuals over the age of 60.	`Correct`	`Correct`
13		...has significantly decreased over the past decade.	`Correct`	`Correct`
14		...is much higher in urban areas compared to rural areas.	`Incorrect`	`Correct`

Note: All test data was equally cleaned as in training with the last step (i.e., reframing questions) made manually. | **Accuracy:** 0.7142